

The Impact of China's DRG Payment Reform on Hospital Incentives and Inpatient Spending

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Abstract

This paper examines the impact of China's Diagnosis-Related Group (DRG) payment reform on hospital behavior and inpatient healthcare spending. The reform replaces the traditional fee-for-service reimbursement system with a prospective payment mechanism that provides hospitals with fixed payments for each diagnosis group. Using province-level panel data and a difference-in-differences framework, we analyze how variation in DRG exposure across regions affects average inpatient cost, inpatient discharge volume, and total inpatient expenditure. The results show that greater DRG exposure is associated with lower average inpatient cost but higher inpatient discharge volume. These two effects partially offset each other, resulting in a small and statistically insignificant change in total inpatient expenditure. The findings suggest that while DRG reform improves cost efficiency per case, hospitals' behavioral responses may limit its ability to reduce overall healthcare spending.

1. Introduction

Healthcare payment systems play an important role in shaping provider behavior and the efficiency of healthcare systems. Different reimbursement mechanisms create different incentives for hospitals regarding treatment intensity, patient admissions, and resource allocation.

China recently introduced a Diagnosis-Related Group (DRG) payment reform that replaces the traditional fee-for-service system with a prospective payment mechanism (National Healthcare Security Administration). Under DRG payment, hospitals receive a fixed reimbursement for each diagnosis group rather than payment based on individual services (Boucher et al. 1). This reform changes hospitals' financial incentives and may affect both treatment decisions and patient admissions.

This paper examines how China's DRG reform affects hospital behavior and healthcare spending. Using province-level panel data and variation in DRG exposure across regions, we estimate the effects of the reform on average inpatient cost, inpatient discharge volume, and total inpatient expenditure. The key economic question is whether DRG reform successfully reduces healthcare spending or whether hospitals respond to the new payment incentives by increasing patient volume. This reflects a classic trade-off in prospective payment systems: while fixed

reimbursement may reduce treatment intensity per case, providers may expand service volume to maintain overall revenue.

2. Reasons for the Policy

DRGs were first developed at Yale University in the early 1970s to group acute-care patients with similar clinical conditions and resource use. In 1983, they became a payment mechanism under Medicare's Inpatient Prospective Payment System. Since then, DRGs have evolved into a broader framework for reimbursement, hospital benchmarking, quality measurement, and cost management (Boucher et al. 2).

Before the introduction of DRG reform, China's hospital reimbursement system relied primarily on fee-for-service payment, which reimbursed hospitals based on the quantity of medical services provided. This structure created incentives for excessive testing, overprescription of drugs, and longer hospital stays, contributing to rising healthcare expenditures (Yip et al. 1). In contrast, DRG payment provides a fixed reimbursement for each diagnosis group, shifting part of the financial risk of treatment costs to hospitals (Busse et al. 146).

China introduced the DRG reform as part of a broader effort to control healthcare spending. The reform establishes fixed reimbursement for each diagnosis group, shifting hospital incentives toward greater cost efficiency. In 2019, the National Healthcare Security Administration launched a pilot program in 30 cities, which was gradually expanded nationwide. By 2023, more than 90 percent of pooled healthcare insurance regions nationwide had begun implementing DRG-based reimbursement systems (National Healthcare Security Administration).

3. Economic Mechanism

The introduction of DRG payment reform changes hospitals' financial incentives. Under the traditional fee-for-service system, hospitals were reimbursed based on the quantity of services provided, encouraging higher treatment intensity and longer hospital stays. In contrast, the DRG system provides a fixed reimbursement for each diagnosis group, hospital profit therefore depends on the difference between DRG payment and actual treatment costs, creating incentives to control costs and improve efficiency.

3.1 Cost Control Incentives

Under DRG payment, hospitals retain any savings when treatment costs fall below the fixed reimbursement, creating incentives to reduce unnecessary inputs and improve efficiency. One common response is to shorten hospital stays.

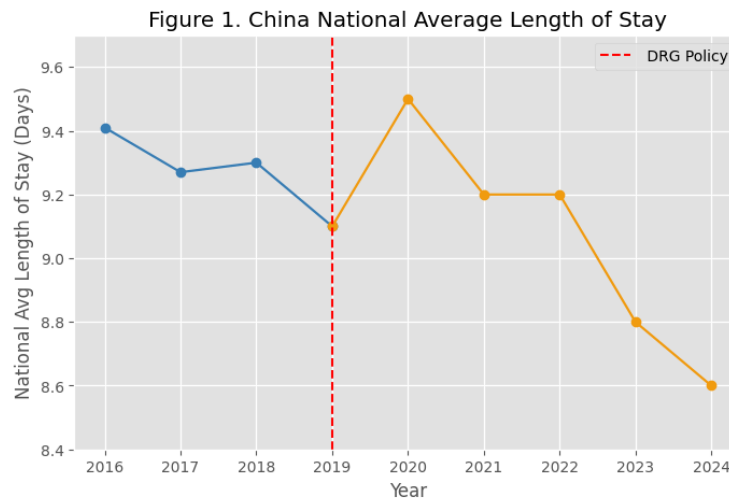


Figure 1 illustrates the decline in the national average length of stay following the introduction of DRG reform.

3.2 Volume Expansion

Although DRG payment constrains revenue per case, it does not directly limit the number of patients treated. Hospitals may therefore increase admissions to maintain overall revenue by improving bed turnover or expanding service capacity. This behavior is closely related to the concept of supplier-induced demand, where healthcare providers influence the quantity of services delivered in response to financial incentives. As a result, patient volume may rise even as the average cost per case declines.

3.3 Patient Selection

DRG payment may also affect patient admission decisions. Because reimbursement is fixed, hospitals may face losses when treating unusually costly patients, creating incentives to avoid

complex or high-risk cases (Silverman and Skinner, 1). This highlights the importance of risk adjustment within the DRG classification system.

Overall, these mechanisms suggest that DRG reform can simultaneously affect treatment costs, patient volume, and admission behavior, shaping hospitals' responses to the new payment incentives.

4. Data

Our analysis uses a province–year panel dataset constructed from official Chinese statistical yearbooks. For each province and year, we collect inpatient discharges and average inpatient cost. Total inpatient expenditure is constructed as:

$$\text{Total inpatient expenditure} = \text{average inpatient cost} \times \text{inpatient discharges}$$

To measure exposure to China's DRG reform, we construct a province-level treatment intensity variable based on policy implementation. Since DRG reform began with pilot cities in 2019, exposure in 2019 is defined as the share of a province's population living in pilot cities. This exposure measure remains constant from 2019 to 2021.

After 2021, the reform expanded nationwide under a three-year rollout plan. Following the policy guideline of 40%, 30%, and 30% expansion, exposure is set to 40% in 2022 and 70% in 2023 for provinces in the expansion phase. Four municipalities—Beijing, Tianjin, Shanghai, and Chongqing—are assigned 100% exposure starting in 2019 because the reform covered the entire municipality. Beijing is treated as fully exposed throughout the sample because it implemented a DRG-type system in 2011.

We include several province-level controls—GDP per capita, aging rate, physicians per 1,000 people, and hospital beds per 1,000 people—to account for differences in economic development, demographics, and healthcare supply. Because expenditure and volume variables are highly skewed, we use logarithms of average inpatient cost, inpatient discharges, and total inpatient expenditure. Data from 2020 are excluded due to distortions from the COVID-19 pandemic.

5. Empirical Strategy

We estimate a difference-in-differences style panel model with province and year fixed effects to evaluate the effect of DRG reform. The main treatment variable is the interaction between policy exposure and a post-reform indicator:

$$\text{Exposure_post} = \text{Exposure} \times \text{Post}$$

where Post equals 1 for years 2019 and after. This captures the effect of DRG exposure after the reform began.

Our baseline specification is:

$$\text{Outcome_it} = \beta \times \text{Exposure_post_it} + \text{Province FE} + \text{Year FE} + \varepsilon_{it}$$

where Outcome_it represents log average inpatient cost, log inpatient discharges, or log total inpatient expenditure. Province fixed effects control for time-invariant provincial differences, while year fixed effects capture nationwide shocks and common trends. Identification therefore comes from within-province changes in DRG exposure after the reform.

We also estimate a specification with additional province-level controls:

$$\text{Outcome_it} = \beta \times \text{Exposure_post_it} + \text{controls_it} + \text{Province FE} + \text{Year FE} + \varepsilon_{it}$$

These controls account for observable factors that may affect both hospital outcomes and reform intensity. However, we treat the fixed-effects-only model as the baseline because it provides a cleaner reduced-form estimate and avoids precision loss in a short panel.

Because total inpatient expenditure equals average inpatient cost multiplied by inpatient discharges, estimating all three outcomes allows us to separate cost effects, volume responses, and the net impact on total spending.

6. Results

6.1 Baseline Result

This section presents the regression results estimating the impact of DRG payment reform on three outcomes: average inpatient cost, inpatient discharge volume, and total hospital

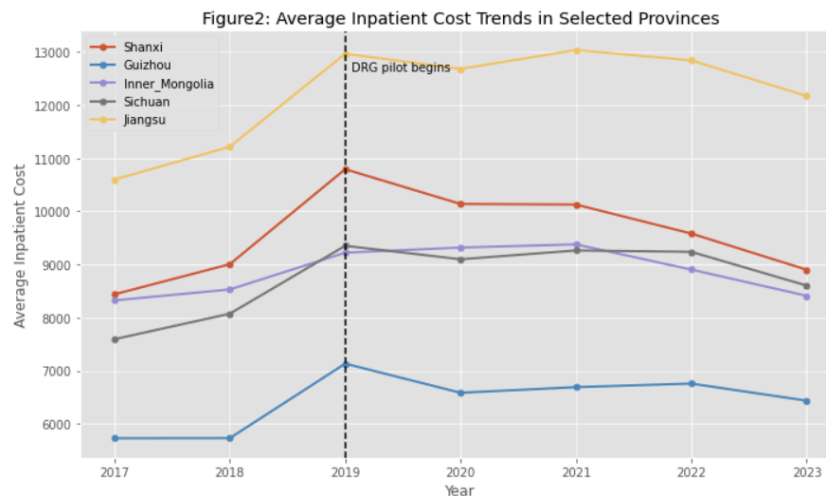
expenditure. The coefficient of interest is the interaction between DRG exposure and the post-reform period.

Results summary:

	coef	std_error	p_value	R_squared	N
Avg Cost	-0.0274	0.0168	0.0498	0.9842	186.0
Volume	0.0455	0.0239	0.0566	0.9969	186.0
Total Expenditure	0.0181	0.0211	0.3920	0.9958	186.0

6.2 Average Inpatient Cost

Figure 2 shows the trend in average inpatient cost before and after the DRG reform in 2019.



The regression results confirm this pattern. The estimated coefficient on DRG exposure is -0.0274 , indicating that provinces with greater exposure to the reform experienced lower average inpatient costs. This result is consistent with the cost-control incentives created by the DRG payment system.

6.3 Inpatient Discharge Volume

The estimated coefficient for inpatient discharge volume is positive (0.0455), suggesting that DRG reform is associated with an increase in the number of treated patients. One possible explanation is that hospitals attempt to offset lower revenue per case by increasing patient volume. Hospitals may therefore increase bed turnover, improve patient throughput, or expand

service capacity to maintain overall revenue. Similar volume responses have been documented after the adoption of DRG-based payment systems in other countries (Busse et al., 2013).

6.4 Total Hospital Expenditure

The estimated effect on total hospital expenditure is positive but small (0.0181) and statistically insignificant. This result reflects the interaction between the previous two effects: while DRG reform reduces the cost per case, the increase in patient volume partially offsets these savings. As a result, the overall impact on total hospital spending appears limited.

6.5 Robustness: Model with Controls

As a robustness check, we estimate an alternative specification that includes additional province-level controls, such as GDP per capita, aging rate, physicians per 1,000 people, and hospital beds per 1,000 people.

Results summary:					
	coef	std_error	p_value	R_squared	N
Avg Cost	-0.0258	0.0175	0.1389	0.9858	186.0
Volume	0.0396	0.0330	0.2306	0.9970	186.0
Total Expenditure	0.0138	0.0314	0.6608	0.9960	186.0

After including these controls, the estimated coefficients remain similar in sign to the baseline specification but become statistically insignificant. This loss of significance likely reflects the limited sample size and reduced variation after including multiple controls. Therefore, we treat the baseline specification as our main result and interpret the controlled specification primarily as a robustness check.

7. Discussion and Policy Implications

The results highlight an important behavioral response to DRG payment reform. While the reform reduces treatment intensity and lowers the average cost per inpatient case, hospitals appear to respond by increasing patient volume. This pattern reflects a classic incentive response to prospective payment systems (Busse et al., 2013). Similar responses were observed after the introduction of the Medicare Prospective Payment System (PPS) in the United States, where

hospitals reduced treatment intensity per case but also adjusted admission behavior in response to fixed reimbursement.

The simultaneous reduction in cost per case and increase in patient volume implies that the overall effect of DRG reform on total healthcare spending may be limited in the short run. Although hospitals become more efficient in treating individual patients, the expansion of service volume partially offsets these cost savings. This result illustrates a broader lesson in health economics: reforms that change provider incentives at the micro level do not necessarily translate into reductions in aggregate healthcare spending.

These findings also have several policy implications. First, policymakers may need to monitor hospital admission behavior to ensure that increases in patient volume do not undermine cost-containment objectives. Second, because fixed reimbursement may create incentives to avoid complex or high-cost patients, strengthening risk adjustment mechanisms within the DRG classification system is important. Finally, policymakers should ensure that cost-control incentives do not compromise healthcare quality by incorporating monitoring mechanisms such as readmission rates or other quality indicators.

8. Conclusion

This paper examines the impact of China's DRG payment reform on hospital behavior and healthcare spending. Using province-level panel data, the results show that greater DRG exposure is associated with lower average inpatient costs but higher inpatient discharge volume.

These findings suggest that although DRG reform improves cost efficiency at the case level, hospitals may respond by expanding admissions, which limits its effect on overall healthcare spending. This highlights an important implication for policy design: payment reforms that change provider incentives may need to be complemented by additional measures, such as monitoring hospital admissions and strengthening risk adjustment mechanisms, to achieve stronger cost-containment effects.

Reference

- Boucher, A., Bowman, S., Piselli, C., & Scichilone, R. (2006). Evolution of DRGs. *Journal of AHIMA*.
<chrome-extension://efaidnbnmnibpcajpcglclefindmkaj/https://journal.ahima.org/Portals/0/archives/AHIMA%20files/Evolution%20of%20DRGs.pdf>
- Busse, Reinhard et al. “Diagnosis related groups in Europe: moving towards transparency, efficiency, and quality in hospitals?.” *BMJ (Clinical research ed.)* vol. 346 f3197. 7 Jun. 2013, doi:10.1136/bmj.f3197, <https://www.bmj.com/content/346/bmj.f3197.long>
- Yip WC, Hsiao WC, Chen W, Hu S, Ma J, Maynard A. “Early appraisal of China's huge and complex health-care reforms”. *Lancet*. 2012 Mar 3;379(9818):833-42. doi: 10.1016/S0140-6736(11)61880-1. PMID: 22386036.
<https://pubmed.ncbi.nlm.nih.gov/22386036/>.
- Silverman, Elaine, and Jonathan Skinner. “Medicare Upcoding and Hospital Ownership.” *Journal of Health Economics*, vol. 23, no. 2, 2004, pp. 369–389.
<https://doi.org/10.1016/j.jhealeco.2003.09.007>.
- National Healthcare Security Administration. *Notice on the Publication of the List of National Pilot Cities for DRG Payment Reform*. 5 June 2019,
https://www.nhsa.gov.cn/art/2019/6/5/art_37_1362.html
- National Healthcare Security Administration. *Notice on Issuing the DRG/DIP Payment Reform Three-Year Action Plan*. 26 Nov. 2021,
https://www.nhsa.gov.cn/art/2021/11/26/art_104_7413.html
- National Healthcare Security Administration. *Policy Interpretation of the Notice on Issuing the DRG/DIP Payment Reform Grouping Plan (Version 2.0)*. 23 July 2024.
https://www.nhsa.gov.cn/art/2024/7/23/art_105_13316.html
- National Bureau of Statistics of China. “National Data: Annual Population by Province.” National Data Database. <https://data.stats.gov.cn/easyquery.htm?cn=E0103>

National Healthcare Security Administration. “*Notice on the Expansion of the DRG Payment Reform Pilot.*” National Healthcare Security Administration, 26 Nov. 2021.
https://www.nhsa.gov.cn/art/2021/11/26/art_104_7413